The Estimator Decides:

What Curricula and Failure Recycling Can and Cannot Do in RL with Verifiable Rewards

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Code, logs, and proofs: https://github.com/linjiw/curriculum-maxrl

Abstract

Post-training with verifiable rewards increasingly relies on two data-level interventions: *difficulty curricula*, which reallocate rollout budget toward learnable prompts, and *failure recycling* (hindsight relabeling), which converts failed rollouts into verified successes of the tasks they actually achieved. Both are treated as objective-agnostic add-ons. We show they are not: the advantage estimator underneath decides whether each helps or harms, in closed form. For success-conditioned (MaxRL-style) advantages, the expected advantage mass a prompt emits from N rollouts is exactly $2(\text{pass@}N - \text{pass@}1)$ — a compute-indexed learnability functional whose band location, width, and scaling follow from the rollout budget alone, and whose two zeros partition difficulty into a dead zone no sampler can reach and a mastered tail where updates buy sharpening instead of signal. Curricula can only reallocate compute within the band between them; recycling is the only channel into the dead zone; and relabels that land in the mastered tail sharpen. This predicts an estimator-conditioned divergence we confirm across four task suites: under identical curricula, success-conditioned runs grow pass@k coverage while GRPO runs lose it. It also predicts a failure mode of recycling we then observe in production — hindsight arms improve mean accuracy while losing coverage, **recycling-induced sharpening** — and supplies the mitigation: gating relabel admission by the destination's pass rate converts the loss into a tunable mean-versus-coverage dial. One diagnosis covers both: audit the estimator before shipping a curriculum or a recycler, and measure coverage, the currency in which both failure modes are visible.

1 Introduction

Post-training language models and control policies with verifiable rewards has a cost structure unlike supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, generation dominates wall-clock (85% of step time, averaged over 272 logged steps across our GSM8K sessions; telemetry committed), and on hard task pools most of that compute buys nothing — under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal**: either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field's response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets [8], target-difficulty controllers [9], category bandits [10], learnability-based rejection sampling [11], dead-prompt filtering [12, 14]. The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently brought into RLVR-style loops for robot policies [18]. Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*: bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

This paper shows that neither assumption survives contact with the estimator's own algebra. The organizing fact is a partition. **The estimator's expected advantage mass $u_N(p)$ divides task difficulty into a dead zone, a learnable band, and a mastered tail. Curricula can only reallocate compute within the band; recycling is the only channel into the dead zone; and relabels that land in the mastered tail buy sharpening, not signal. The estimator decides all three.** Three questions unpack this:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected advantage mass a task emits from a group of N rollouts has a closed form — $2(\text{pass@}N - \text{pass@}1)$, a compute-indexed learnability functional (§3). Its two zeros are the partition's boundaries. Mass is a *surrogate* for learning signal (Remark 1), but it is the part a sampler can act on before spending a rollout, and it makes the allocation *derivable* rather than heuristic: the band's location, width, and N-scaling come from the budget you already chose, and published learnability curricula are the $N{=}2$ slice. It also bounds the channel. Allocation can only redistribute signal that exists: a perfect-information allocator ties our cheap posterior in end performance (§6.1), and at LLM scale a prompt-level teacher starves before it pays (§6).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N{-}1)\times$ more mass than RLOO on hard tasks as $p \to 0$, and releases mastered tasks; GRPO's variance-normalized weights flatten the profile and keep spending on both tails. We predict and measure the resulting divergence at two scales: on the maze, every MaxRL run grows pass@8 coverage and every GRPO run loses it, across four teacher conditions (§6.3); on GSM8K, the pre-registered prediction that GRPO+teacher would be the only regressing cell landed on the registered run, with the replication status reported in §6.7. Recycling has its own, previously unreported failure mode, and the partition predicts it: relabels that land in the mastered tail sharpen. On Countdown arithmetic this is **recycling-induced sharpening** — exact-verifier relabeling improves mean accuracy while *losing* pass@k coverage (§6) — the data-induced sibling of GRPO's weight-induced collapse, formally the composition of hindsight's marginal-distribution shift [16] with curation-driven self-consumption [17].

Q3: Does the algebra also supply the fix? Yes — by applying the functional a third time. A relabel to a destination the model already reaches sits in the mastered tail, where mass is zero, so we gate relabel admission by the destination's estimated pass rate: the same utility that schedules sampling, now deciding what the recycler may admit. Across three seeds the gate restores frontier coverage to the no-recycling baseline while keeping most of recycling's mean gain, and sweeping its strength traces a monotone mean-versus-coverage trade — a dial, not a switch (§6.9). Where recycling is healthy the gate is transparent: hard-destination relabels pass untouched, so the validated testbeds reproduce their results with the gate enabled. One method, no per-domain switches.

Our contributions:

1. **A closed-form curriculum** (§3): the expected advantage mass of a prompt under success-conditioned RL is exactly $2(\text{pass@}N - \text{pass@}1)$, a compute-indexed learnability functional whose band, width, and N-scaling are derived rather than tuned, and whose zeros partition difficulty into the dead zone, the band, and the mastered tail; published learnability curricula are its $N{=}2$ slice.

2. **A derived teacher and a characterized recycler** (§4–5): FrontierMax schedules prompts by this functional with the per-group estimator untouched, and hindsight relabeling under a verified-success contract yields the ML gradient of the achieved task under the original sampling law.

3. **Estimator-conditioned predictions, confirmed** (§6): under identical curricula, success-conditioned runs grow pass@k coverage while GRPO runs lose it, across four task suites from exact-gradient chains to LLM post-training.

4. **A new failure mode and its derived fix** (§6.8–6.9): exact-verifier recycling improves mean accuracy while losing coverage — recycling-induced sharpening — and gating relabels by destination pass rate converts the loss into a monotone mean-versus-coverage dial.

Along the way we state a refinement to the MaxRL paper — the practical drop-$K{=}0$ estimator is unbiased for the $T{=}N{-}1$ objective, not $T{=}N$ (Lemma 1) — and every prediction below was committed before the deciding runs, with negatives and retractions reported in place (§8) and audited in the repository.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1-p)$ and truncates at order T, giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE ($T{=}1$) to exact ML ($T \to \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped (Lemma 1 gives its exact truncation order). Its weight function grows as $p \to 0$: the

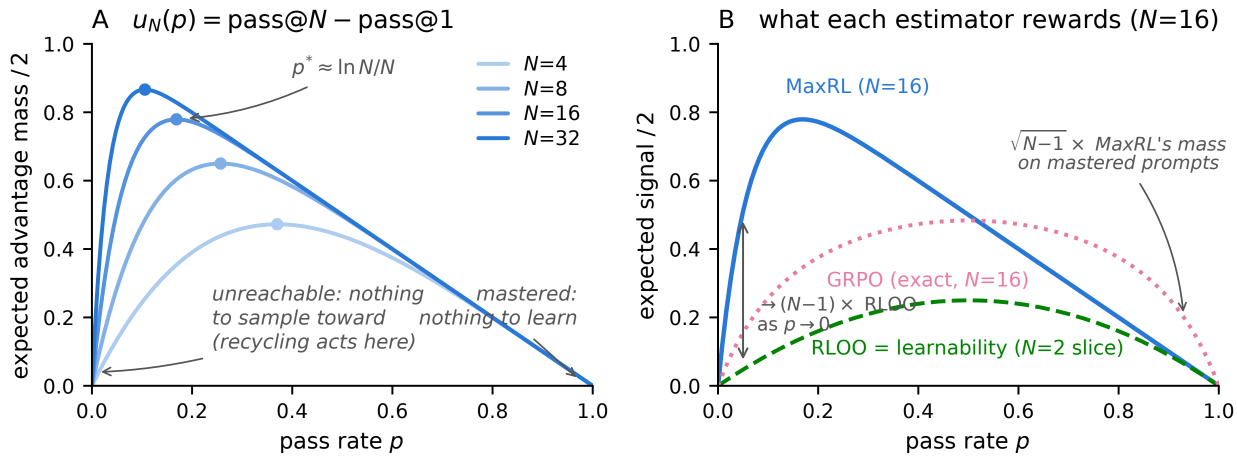

Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1-(1-p)^N) - p$ — the estimator's expected advantage mass (Prop. 1) — for growing rollout budgets N. The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; both tails are dead zones the weight function cannot reach. *Right:* all three curves are exact expected mass at $N{=}16$ (GRPO's is $\frac{1}{N}\mathbb{E}\sqrt{K(N{-}K)}$, $K \sim \text{Bin}(N,p)$; MC-verified). MaxRL concentrates $\to (N{-}1)\times$ more mass than RLOO on frontier prompts (Prop. 2) and releases mastered prompts fastest; GRPO's variance normalization keeps $\sim\sqrt{N{-}1}\times$ MaxRL's relative mass on mastered prompts — the maintenance-of-easy-tasks its curriculum arms lose in §6. RLOO's profile is exactly the published "learnability" objective — the $N{=}2$ slice.

objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

3 The estimator is the curriculum

Lemma 1 (Truncation order of the practical estimator)**.** *The drop-$K{=}0$ variant of the practical MaxRL estimator is exactly unbiased for the $T{=}N{-}1$ member of the objective family, not $T{=}N$: zeroing the control variate only on all-fail groups makes it outcome-dependent, contributing $-(1-p)^{N-1}\nabla p$, and $\sum_{k=1}^{N}(1-p)^{k-1} - (1-p)^{N-1} = \sum_{k=1}^{N-1}(1-p)^{k-1}$.*

Interpretation. *A refinement to Tajwar et al. [1], immaterial for their conclusions but load-bearing for our mass accounting at small N — all identities below are for the estimator as actually shipped.*

Proposition 1 (Advantage mass)**.** *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights,*

$$\boxed{\mathbb{E}\Big[\textstyle\sum_i |w_i|\Big] \;=\; 2\big(\textit{pass@}N(p) - \textit{pass@}1(p)\big) \;=\; 2\big((1-(1-p)^N) - p\big) \;\equiv\; u_N(p).} \tag{1}$$

Proof (three lines): on $\{K \geq 1\}$, $\sum_i |w_i| = K(\frac{1}{K} - \frac{1}{N}) + \frac{N-K}{N} = 2(1-\frac{K}{N})$, and $\mathbb{E}[2(1-\frac{K}{N})\mathbf{1}\{K \geq 1\}] = 2(\Pr[K \geq 1] - p) = 2(\textit{pass@}N - \textit{pass@}1)$.

Interpretation. *The estimator's expected learning signal on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. That sentence is a curriculum, and its two zeros are a* ***partition of difficulty****:* a dead zone *($p \to 0$: nothing to condition on), a* learnable band *peaking at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$, and a* mastered tail *($p \to 1$: no contrast left). The band's center, width, and compute-scaling are set by the one parameter you already chose. Everything in this paper is one of three moves against this partition: allocate within the band (§4), create signal in the dead zone (§5), and refuse updates that land in the mastered tail (the gate).*

Proposition 2 (Frontier amplification). *As $p \to 0$, the ratio of MaxRL's expected signal to RLOO's ($2p(1-p)$ exactly) tends to $N-1$.*

Interpretation. *The comparison to GRPO is two-sided and exact: GRPO's expected mass is $\frac{1}{N}\mathbb{E}\sqrt{K(N-K)}$ with $K \sim \mathrm{Bin}(N,p)$, and taking tails, MaxRL/GRPO $\to \sqrt{N-1}$ as $p \to 0$ while GRPO/MaxRL $\to \sqrt{N-1}$ as $p \to 1$. MaxRL over-serves the frontier and releases mastered prompts; GRPO silently keeps $\sqrt{N-1}\times$ more of its mass on mastered prompts — maintenance a curriculum then removes. Concentrating sampling on the frontier only helps if the estimator extracts signal there, and §6 shows the mastered-tail difference becoming an active failure under a curriculum.*

Proposition 3 (Allocation). *The rollout allocation maximizing total expected advantage mass is greedy water-filling on the marginal $p(1-p)^N$ — the probability that the next rollout is a group's first success.*

Interpretation. *This is the mass-optimal* reference *allocator, not the deployed sampler: the practical teacher (§4) keeps fixed group sizes and samples prompts $\propto u^\gamma$, trading optimality for group-parallel batching and for the exploration a point-optimal allocator forfeits (the unconstrained maximizer concentrates on the current arg-max, a brittle choice under posterior noise). We measure what the gap costs: a true-pass-rate water-filling oracle ties the Thompson teacher in end performance (§6.1).*

Remark 1 (Scope of the surrogate). Advantage mass $\mathbb{E}\sum_i |w_i|$ is the model-free factor of the policy gradient $\sum_i w_i \nabla \log \pi(z_i)$: it counts weighted rollouts, not the score-vector geometry (norms, alignment, cancellation) that also shapes the realized update. We use it as the sampling utility for three reasons. (i) It is what a sampler can know *before* spending rollouts — score geometry is only observable after generation, when the budget is already spent. (ii) Its zeros are exact: at $p \in \{0,1\}$ the estimator emits no gradient of any norm, so the dead zones it predicts are real regardless of geometry. (iii) On the exact-gradient testbed, where the true first-order expected eval improvement per group has a closed form, mass *ties* it as a ranking of tasks (Spearman within .005 at every training snapshot; artifact `results_bridge.json`) — as a predictor of the next step, u_N is not missing anything material. What mass does not claim to be is *the* optimal sequential utility, and the same artifact shows why no state-local functional is: on a pool of independent tasks the true final-performance optimum is a *planning* solution (a trajectory-knapsack allocation with a budget-dependent funding cutoff), and which surrogate best approximates it depends on the training objective — a variance-tilted $(1-p)^\alpha u_N$ wins time-averaged (anytime) performance while model-predictive replanning wins the final checkpoint, each losing the other's metric (8–10/10 paired seeds each way). "Best curriculum utility" is ill-posed until that objective is fixed; we report both currencies throughout. §6.5 shows the regime where the whole channel buys nothing (all tasks learnable at budget), and the partition's boundaries — the load-bearing claim — are the zeros all these candidates share.

Remark 2 (The curriculum changes the training distribution). Sampling prompts from an adaptive q_t instead of the data distribution ρ optimizes a reweighted objective $\mathbb{E}_{x\sim q_t}[J(x)]$: each group's gradient is an unbiased MaxRL gradient *for its own prompt*, but the mixture is tilted toward the frontier, without importance correction back to ρ. This is the standard bargain every curriculum strikes (uniform sampling makes the same choice for $q_t{=}\rho$), and we treat it as an empirical question, not a free lunch: all evaluations in §6 are on fixed held-out pools under ρ, so any mismatch cost is charged to the method. The safety results are precisely about when this tilt helps or hurts the ρ-measured outcome.

One identity, three literatures: learnability curricula [11, 25] are the $N{=}2$ slice of u_N; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling [15] is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). Closest prior: DisCO [2] derives the per-question weights of the GRPO family ($\omega(q) = \sqrt{p(1-p)}$ for GRPO, $p(1-p)$ for Dr. GRPO) to diagnose difficulty bias, and SC-SDPO [3] notes the residual $\sqrt{p(1-p)}$ acts as an implicit curriculum. Our identity generalizes this line to success-conditioned estimators (where the functional is $u_N = \text{pass@}N - \text{pass@}1$, compute-indexed by N), and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

The partition, as a map of the paper (Fig. 3). The **teacher** (§4) allocates compute within the band — and saturates: a true-pass-rate oracle allocator ties the cheap posterior in end performance (§6.1), so better pass-rate information is not where remaining gains live. **Recycling** (§5) creates signal in the dead zone —

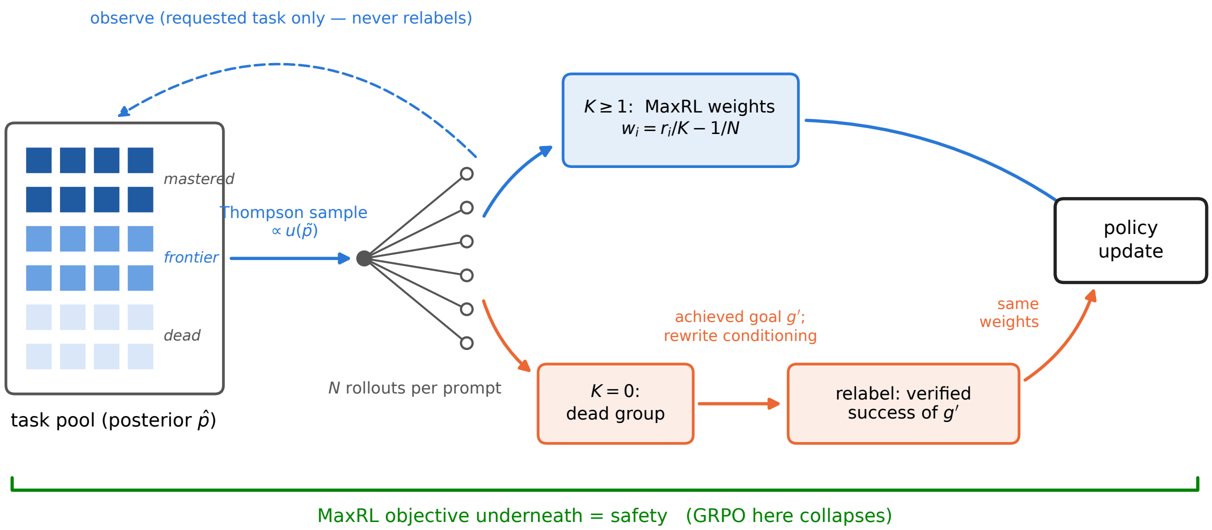

Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

Algorithm 1 FrontierMax: one step. Three lines differ from a standard group-rollout loop (marked ▷).

▷ **sample** prompts $x_1..x_B \propto u_N(\tilde{p}_x)^\gamma$ mixed with a uniform floor, $\tilde{p}_x \sim \text{Beta}(\alpha_x, \beta_x)$ (else: uniform)
roll out N completions per prompt; verify; $K_x \leftarrow$ successes
update posterior (α_x, β_x) from (K_x, N) *for the requested task only*
live groups $(1 \leq K_x)$: MaxRL advantages $w_i = r_i/K - 1/N$, unchanged
▷ **dead groups** (K_x=0): relabel to the achieved sub-goal g' (verified; conditioning rewritten)
▷ admit the relabel only if $\hat{p}_{g'} \leq$ `gate_max_p` (the gate: refuse the mastered tail)
one policy-gradient step on live + admitted relabeled groups

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the only channel that still adds on top of the oracle. The **estimator** underneath decides whether either is safe, and its mastered tail is where both failure modes live (§6). One line: *the teacher allocates, hindsight creates, the estimator decides whether either is safe.*

4 FrontierMax: the schedule

FrontierMax is a standard group-rollout RLVR loop with three modified lines (Alg. 1). A decayed Beta posterior (decay 0.7) tracks each prompt's pass rate from observed group outcomes only — never from relabeled successes (violating this inflates the posterior: $\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a 10% uniform floor; $\gamma \approx 4$ when tasks share skills (learning compounds — validated on chains, and its non-transfer to broad pools was predicted in advance by a compounding model), $\gamma = 1$ otherwise. Live groups train with unmodified MaxRL advantages. Knob inventory, stated plainly: decay, floor, and γ exist, with validated defaults; what is *derived* rather than tuned is the difficulty band itself — location, width, and N-scaling — which is exactly where competing curricula spend their hyperparameters.

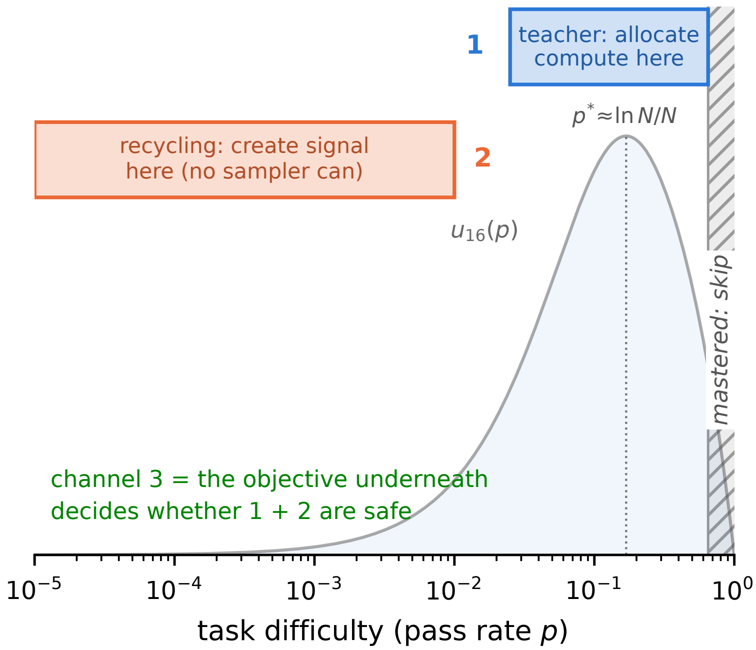

Figure 3: **The partition as a map of the method.** The teacher allocates compute within the learnable band; recycling creates signal in the dead zone no sampler can reach; the gate refuses relabels that land in the mastered tail; the estimator underneath decides whether any of it is safe.

5 Failure recycling

Proposition 4 (Hindsight update, characterized)**.** *Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields the ML gradient of task g' under the conditional law induced by the* original *task's sampling; it equals the fresh-rollout gradient exactly when the two conditional laws match, and is otherwise a distribution-shifted (biased) estimate of it.*

Interpretation. *So "exactness" is a property to* measure per environment*, not a blanket guarantee. On our validation testbed the laws match by construction (prompt-independent skill execution), so cosine probes there are consistency checks, not evidence — internal review showed several null relabeling schemes produce the same cosines on that testbed, and we do not cite them as support. The evidence that the relabel* direction *carries information is behavioral instead: the placebo decomposition of §6.1 (*+0.037 *beyond dose-matched replay, the tightest arm in the battery; random-target relabels lose to replay 5/5 seeds). In Countdown the relabel keeps the trajectory's own achieved value — the shift enters only through which expressions get attempted for which targets — and the verified-success guarantee (contract 1) holds regardless. What the shift costs in general is an open, environment-specific question.*

Two contracts keep practice on the proof's side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment's own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Recycling is the only mechanism here that *creates* signal, and we decompose its headline gain rather than let it inflate: of the +0.22 AUC on fixed task pools, a placebo replaying *live* gradients on dead-group slots captures 83% — most of the benefit is extra gradient dose on otherwise-wasted slots — while the relabel *direction* carries +0.037 beyond the strongest placebo and is exactness-dependent (random-target relabels are harmful). Both numbers ship together. The value is regime-dependent in a way we can state precisely: the gain is proportional to how much a relabeled skill can compound — +0.22 AUC on fixed task pools, +0.01 on one-shot task streams. Task-set fixedness is the single most important regime variable we found.

Table 1: The ladder at a glance: every rung asks one question of the partition; scale rises, control falls.

rung	question	setting	seeds	outcome
6.1	does allocation saturate?	skill chain (exact grads)	5	yes: oracle ties posterior
6.2	anything to sample toward?	frontier-heavy pool	5	no: only creation scores
6.3	main effect, real gradients?	maze transformer, 1.26M	9 runs	yes: 9/9, p=0.0079
6.4	where/when does coverage go?	maze, level-resolved	22 runs	easy band, immediately
6.5	when does allocation not pay?	IsaacLab Anymal-C	1	learnable-everywhere: null
6.6	beyond task spread?	MountainCar/CartPole	3	spread=existence; teacher=speed
6.7	LLM scale?	GSM8K 2×2, 360M	1–2	sign landed; replication open
6.8	what does recycling cost?	Countdown v2	3	mean up, coverage down
6.9	does the gate recover it?	Countdown v2	3	yes: a monotone dial

6 Experiments: an escalating ladder

Each rung asks one question of the partition, at increasing scale and decreasing experimental control; Table 1 is the map. Protocol, stated once: ± denotes standard deviation across seeds; paired contrasts share per-seed warmstarts; permutation tests are exact (all label assignments); "coverage" is pass@k at the k named per suite (8 on the maze, 16 on Countdown, 4 on GSM8K); AUC is the mean of the eval curve over training at matched wall-clock or episode budget. Pre-registered predictions were committed before the deciding runs; negative results appear in place, one sentence each, with full postmortems in the repository's audit trail.

6.1 Does allocation saturate? (exact-gradient skill chains; CPU, 5 seeds). Uniform 0.650 → teacher 0.728 → full stack **0.890 ± 0.002** AUC. The control battery locates where that comes from, and its eight arms order into a clean *gradient-information ladder*: zero-information dose (lr×2 0.798 ≈ random-target relabels 0.800) < reused correct direction (live-gradient replay 0.832) < new correct information (true relabels 0.869) < perfect allocation (oracle 0.8885 ≈ full stack 0.8895) < allocation+creation (oracle+recycling 0.8935) — each rung a 5/5 paired-seed win. *Allocation saturates:* the floor- and γ-matched true-pass-rate oracle ties the full stack; perfect information about pass rates is worth almost nothing over the cheap posterior. *Creation decomposes:* the replay placebo captures 83% of recycling's +0.22; the relabel direction carries +0.037 beyond it — and the direction term is the most *reliable* arm in the battery (seed spread ±.0025, 4–6× tighter than every placebo; its worst seed beats every placebo's best). Precision on the converse: random-target relabels are harmful *relative to dose-matched replay* (−.032, 5/5 seeds) but ≈ neutral vs. plain lr-doubling — fake labels waste the slot rather than poison training. Utility-form choice, same protocol: exact u_N 0.728±.002 ≈ legacy frontier form 0.733±.013 (the derived form's payoff is 7× smaller seed variance, not mean speed), but the N-blind N=2 learnability slice $p(1-p)$ forfeits the *entire* allocation gain — paired against uniform sampling it is a wash (−.007 mean, 2/5 wins) where u_N carries +.078 (5/5): the group size in the utility is load-bearing. At the endpoint: all arms finish within .027 of each other at convergence — on fixed pools these interventions buy *time-to-ceiling*, not the ceiling (AUC spread 9× the final spread), which is exactly the currency — rollouts — that costs money at LLM scale.

Takeaway. A monotone information ladder: dose < direction < allocation-ceiling < +creation. Allocation saturates; creation adds on top; and every intervention buys speed, not asymptote — efficiency is the currency that prices them.

6.2 Can any sampler act when nothing is learnable? (frontier-heavy regime, max pool pass rate 10^{-5}; 5 seeds, artifact committed). Uniform, DAPO, and the plain teacher score **exactly 0.00**; adding recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ∼400 groups, then goes silent. The attribution control pins down which channel did it: *uniform*+recycling scores the same (0.931±.002 vs. teacher+recycling 0.928±.003 AUC) — in this regime the entire effect is signal creation; allocation contributes nothing because there is nothing learnable to allocate toward. Equivalently: a designer who hand-added the achieved sub-goals to the pool would also escape the zero — what recycling buys is *discovering* that sub-pool from the policy's own failures, with no task engineering.

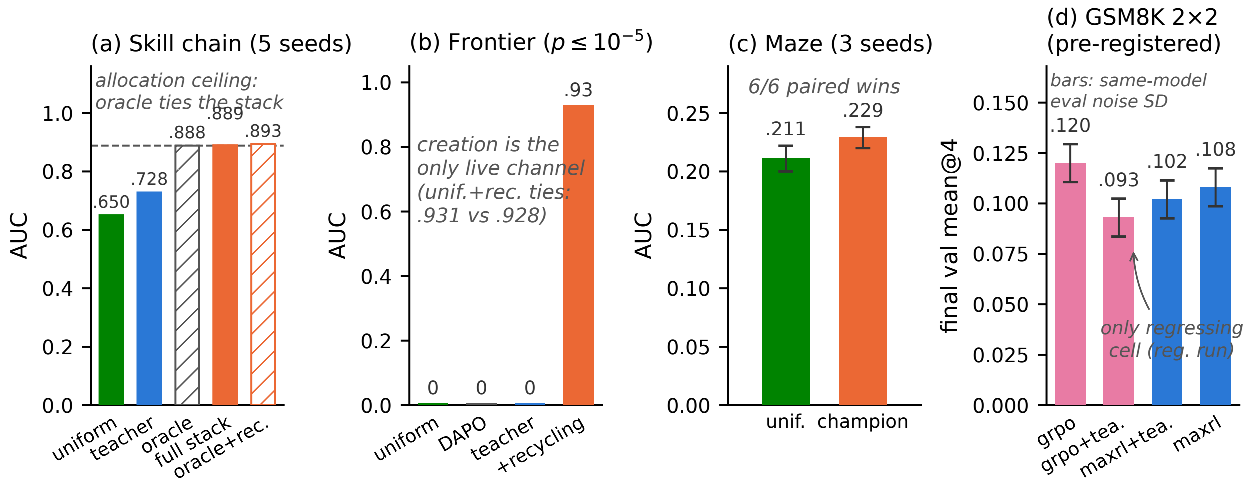

Figure 4: **The ladder.** (a) Exact-gradient chains: a floor-and-γ-matched true-pass-rate oracle *ties* the full stack — allocation saturates — and recycling still adds on top of the oracle. (b) Frontier-heavy regime: allocation scores exactly zero; recycling solves it, and does so equally under uniform sampling (.931 vs. .928) — creation, not allocation, is the live channel. (c) Maze at matched wall-clock: 6/6 paired seed wins. (d) GSM8K 2×2, registered run, error bars = measured same-model eval noise (SD .0094): GRPO+teacher is the only cell that regresses (pre-registered P-G2; a replication seed did not reproduce it — §6.7); the teacher's MaxRL effect is a null at this budget (P-G1).

Takeaway. Where nothing is learnable at budget, sampling policy is irrelevant — uniform and teacher tie at zero without recycling and tie at 0.93 with it. Creation is the only live channel, and its value is discovery, not allocation.

6.3 Does the estimator main effect survive real gradients? (maze transformer at matched wall-clock; 1.26M params, 13 levels, ∼30 runs, 3 seeds). Champion (teacher + dense recycling) 0.252±0.005 final / 0.229±0.009 AUC vs. uniform 0.230±0.015 / 0.211±0.011; paired deltas positive 6/6 seeds; 22–35% more optimization steps per GPU-hour. Attribution within that gain, stated plainly: matched *wall-clock* lets curriculum arms take 1.2–1.4× more optimization steps (dead groups skip the backward pass), and step-matched truncation shows the extra steps are most of the pure teacher's effect (step-matched teacher-only ΔAUC +0.006, p≈0.36, n.s.) while teacher+recycling survives step-matching (p≈0.04) — on the maze the teacher's measured value is *throughput* (buying more steps per GPU-hour), recycling's is signal. The maze teacher uses the legacy frontier form; a pre-registered 3-seed rerun with the exact u_N measures the two within noise *on the maze itself* (AUC .2138±.0059 vs. .2135±.0104; artifact `un_form_verdicts.json`), closing the gap between the deployed form and the derived one. The same sweep refutes a CPU-derived within-band refinement (a $(1-p)$-tilted utility that won 10/10 on the testbed loses 0/3 on the maze on every meter) — within-band utility choices are testbed-sensitive, which is why the theory claims only the band's boundaries. Safety, stated as what the design supports: an estimator *main effect on coverage* — all five MaxRL runs grow pass@8 (e.g. 0.316 → 0.348 under the teacher) and all four GRPO runs lose it (0.308 → 0.271 uniform; 0.332 → 0.269 with the teacher), permutation p=0.0079 on the 9-run main effect. Whether the teacher *amplifies* GRPO's decay is a factorial (interaction) question this suite is not powered for; the GSM8K 2×2 below tests exactly that cell, pre-registered. Inference currency, one convention stated and applied to every level (smallest k reaching absolute coverage 0.25, log-interpolated; single checkpoint pair, artifact `maze_gpu/efficiency.json`): our checkpoint needs **11**× fewer samples at level 5, ties GRPO at levels 2–3, and needs 3× more at level 4 — the frontier is where the efficiency gain lives, and the ledger is mixed below it. The summed-coverage view says the same thing without a target choice: GRPO ahead at pass@1 (3.24 vs. 3.07 summed over L0–6), curves crossing at k≈4, MaxRL ahead by half a level at k=64

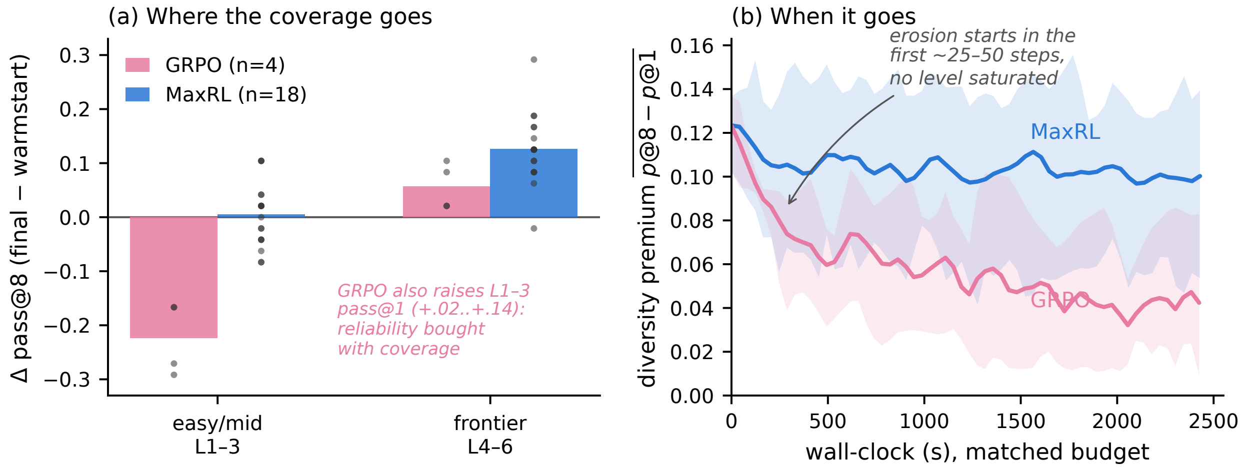

Figure 5: **Where and when GRPO's coverage goes (maze, level-resolved, matched wall-clock).** *(a)* Δpass@8 (final − warmstart) by difficulty band: GRPO's loss lives on the easy/mid band (all 4 runs; dots are runs) while its pass@1 there rises — sharpening; MaxRL holds the easy band and gains on the frontier band (18 runs; band asymmetry exact permutation p=0.0001). *(b)* Diversity premium $\overline{\text{pass@8} - \text{pass@1}}$ vs. wall-clock (mean and min–max across runs): GRPO's erosion begins within the first ∼25–50 steps, before any level saturates; MaxRL stabilizes after warmup.

(5.12 vs. 4.62) with GRPO's level-2/4 curves saturating at 0.88/0.56 where ours reach 1.00/0.62 — choose the estimator by how many samples you will draw at inference.

6.4 Where and when does GRPO's coverage go? (level-resolved). Decomposing the main effect by difficulty band turns it into a mechanism (Fig. 5). *Where:* GRPO's pass@8 loss is concentrated almost entirely on the easy/mid band L1–3 (−.224 mean over its four runs, every run negative) while its L1–3 pass@1 *rises* (+.02 to +.14, every run) — it funds reliability by liquidating easy-band coverage; MaxRL holds L1–3 flat (+.005±.058 over 18 runs) and concentrates its gain on the frontier band L4–6 (+.126±.063 vs. GRPO's +.057). The band asymmetry is exact-permutation p=0.0001. This is the observable consequence of the mass curves in Fig. 1: the $\sqrt{K(N-K)}$ weighting keeps spending updates on mastered prompts (sharpening them), where u_N has already released them. *When:* the divergence is immediate, not saturation-triggered — GRPO incurs 59–73% of its total diversity-premium loss ($\overline{\text{pass@8} - \text{pass@1}}$: .123 → .042) within the first quarter of the budget, with the first easy-band pass@8 drop visible at steps 0–50 when L1 pass@1 is still only .62–.75; MaxRL's premium stabilizes after warmup (.122 → .101). A dose-response note on recycling from the same logs: sparse relabeling (1/dead group) is the only variant that preserves the diversity premium (Δgap −.005±.009 vs. −.030 for both no-recycling and dense; exact p=0.007/0.008), while dense banks the same coverage as sparse into the best final pass@1 — on the maze, dose sets where the coverage dividend is spent, not whether coverage is lost (final pass@8 across doses: null, p=0.39).

Takeaway. The gains survive real gradients, and the coverage main effect now has a located mechanism: GRPO liquidates easy-band coverage for reliability from the first updates; MaxRL reallocates the same budget to the frontier. Coverage is the currency that sees it.

6.5 When does allocation *not* pay? (IsaacLab legged locomotion, pre-registered null). A 5-arm pilot on Anymal-C rough terrain (frontier teacher vs. stock greedy walker, uniform, scripted ramp, and a no-motion control; fixed-grid per-level evaluation, one seed, run by an independent team on a co-designed protocol) lands the *pre-registered* null exactly: on a terrain grid where every level is learnable at budget, the stock greedy curriculum wins outright (mean pass .372 vs. the teacher's .278), and the teacher's only edge is a within-noise sliver at the hardest evaluated level. Combined with the maze step-matched analysis and

Countdown, this makes the regime rule a three-scale result: **allocation pays only where unlearnable-at-budget regions exist to avoid** — on learnable-everywhere pools, breadth beats focus at 1.26M-param maze, 360M LLM, and legged-robot scale.

6.6 What does a curriculum add beyond task spread? (Gymnasium control — binary-success RLVR on classic control, official-task currency. Setting, stated precisely because the baseline label matters: all arms train on *binary task success* (MountainCar: reach $x \geq x^*$; CartPole: survive $\geq T$) with the same group estimator — the classic-control analogue of a verifier reward, not the environments' native dense rewards that standard libraries train on. The **target-only** arm (sample only the official task) is then a provable zero-update freeze, not a slow learner: every group is all-fail, every group-relative estimator (MaxRL, GRPO, RLOO) emits exactly zero weights on constant rewards, and telemetry confirms the policy received *zero* parameter updates over its full budget ($\|\theta\| = 0$ throughout; measured random-policy success is $< 3{\times}10^{-6}$ for the flag over 10^6 episodes and $\approx 10^{-13}$ for survive-400 by tail fit — no budget ignites it). The gated full stack under the identical reward reaches **1.000±0.000 on both official tasks** (3 seeds, Fig. 6). The dense-reward world, measured rather than assumed (prediction registered before the run): with CartPole's native +1/step reward, vanilla REINFORCE on the *same tile policy* solves survive-400 (0.96±0.03 at plateau, 3 seeds) — the categorical zero is a property of sparse verifier-style reward, which is exactly the RLVR setting this rung models; with MountainCar's native reward (−1/step, constant until first success) the same dense baseline stays at 0.000 in all seeds over its full budget, which is why it is *the* hard-exploration classic. Attribution within the sparse setting, three rungs at a common 280-step horizon (hard-task AUC): uniform-over-bins task spread gets most of the way (MC .750 → +recycling .895 → +teacher .956; CartPole .708 → .792 → .893; the paired teacher increment is positive in 3/3 seeds on both envs), creation-without-allocation plateaus *below* ceiling on CartPole (0.958 in every seed) while adding the teacher restores 1.000, and the teacher's remaining contribution is speed and stability: 3.0× faster to hard-task 0.9 on both environments (2.1× to 0.99 on CartPole) with 2.5–6× smaller cross-seed variance along the whole curve. An instructive methods note: our own first version of this study silently used per-bin task-partitioned policy parameters and reproduced the known failure (flag 0.000 in every arm) — restoring the validated shared-policy design (the task enters only the success predicate) is what cracks the task, replicating the 10-seed transition-matched branch study (flag 0.848±0.058) and strengthening it to full convergence.

Takeaway. Curricula operate through shared parameters, or not at all.

6.7 Does it transfer to LLM scale? (GSM8K 2×2, pre-registered; (SmolLM2-360M, N=16, one A10G). Predictions committed before any cell finished. **P-G2 landed on the registered run:** GRPO+teacher was the only cell that regressed in its second half, and the deficit against plain GRPO at the endpoint is 2–4× the measured same-model eval noise (mean@4 −.027, $z = -2.0$; pass@4 −.048, $z = -2.0$; the second-half dip itself, .096 → .093, is within the ±.009 noise floor and carries no weight on its own) while plain GRPO climbed monotonically (.078 → .105 → .120). Steering telemetry, honestly read: the teacher's sampled-dead fraction *minimum* reaches .48 vs. uniform's .52, but the run means are indistinguishable (.66–.67 in all cells) — the delivered treatment was weak, which is the posterior-starvation mechanism below, and it bounds how much of the GRPO gap the curriculum can own. **A replication seed does not reproduce the regression shape:** seed 2 of the same cell climbs monotonically (.073 → .095 → .118), and its steering telemetry shows the treatment was weaker (min dead-sampled fraction .53, mean .66 ≈ population — a near-uniform teacher on average). The completed *within-seed* pair (grpo-uniform seed 2: .081 → .085 → .125, final pass@4 .229; artifact `grpo_uniform_seed2.json`) sharpens the read: the endpoint teacher deficit under GRPO has the same *sign* in both seeds on both meters (4/4 signed contrasts; seed 2: −.007 mean@4, within noise, −.026 pass@4, ∼1.5× its floor) with magnitude tracking steering intensity — a dose effect, not a sign flip. The pre-registered *second-half-regression* shape remains 1 of 2 seeds: suggestive, not established. MaxRL+teacher trains stably to its best value (.066 → .102); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for MaxRL) a *null at this budget* — predicted in advance by the posterior-starvation analysis below. A pass@k sweep on the registered checkpoints: the teacher's deficit under GRPO *widens monotonically with* k (−.008 at k=1 to −.024 at k=16) — curriculum damage lives in coverage, as the maze predicted, though each individual delta is within single-seed noise (and the sweep's absolute level sits ∼3× below the trainer's val metric, an unreconciled harness discrepancy; only the within-sweep contrast is quoted).

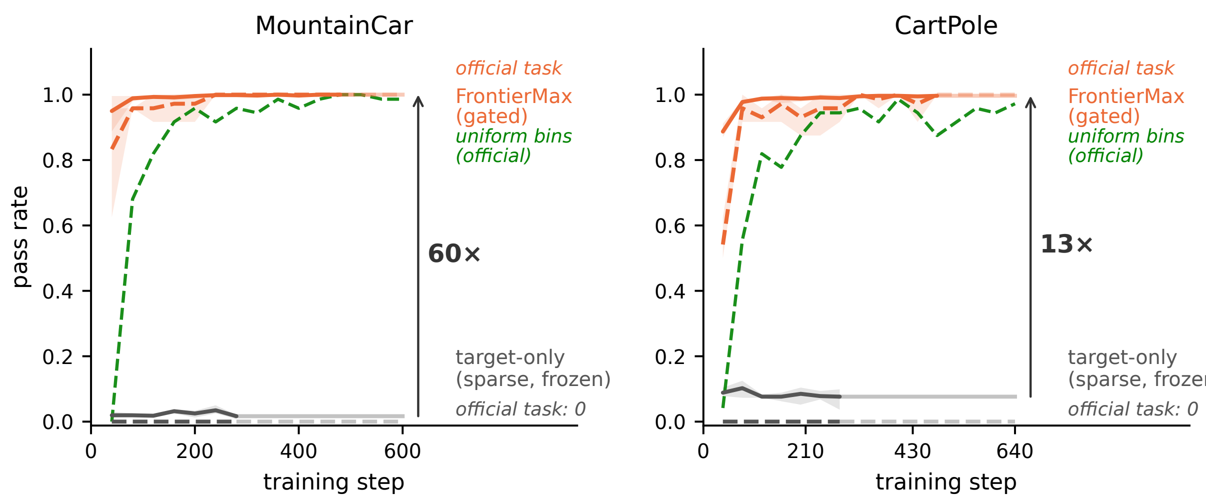

Figure 6: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). The target-only arm (gray) is a provable zero-update freeze under sparse verifier-style reward — all-fail groups emit zero gradient in every group-relative estimator (with CartPole's native dense reward, plain REINFORCE solves the task; the zero is a property of the sparse setting this rung models). A uniform-over-bins control (green) also cracks the task — 3× slower to hard-task 0.9 and off-ceiling — isolating the teacher's contribution (speed, seed-stability) from task spread's (existence of the gradient); the gated FrontierMax stack converges to 1.000±0.000.

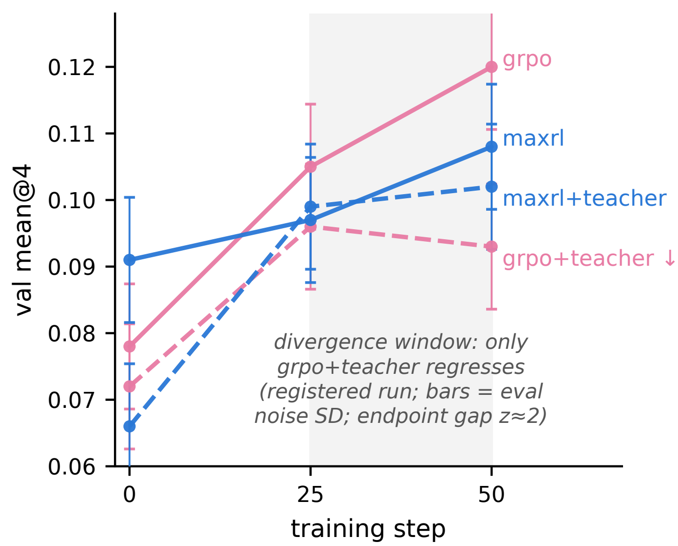

Figure 7: **GSM8K 2×2 (pre-registered; registered run shown; error bars = measured same-model eval noise, SD .0094).** GRPO+teacher is the only cell that regresses in its second half (P-G2 on this run; the load-bearing statistic is the endpoint gap vs. plain GRPO, $z \approx 2$ — the second-half dip alone is within noise; a replication seed with weaker measured steering climbed instead — §6.7); under MaxRL the identical teacher trains stably to its best value but does not beat uniform at this budget (P-G1: null).

One design confound is disclosed rather than dismissed: the teacher samples with replacement, so curriculum cells revisit prompts more (∼34% unique prompts seen vs. ∼43% uniform) — a repetition effect on GRPO is

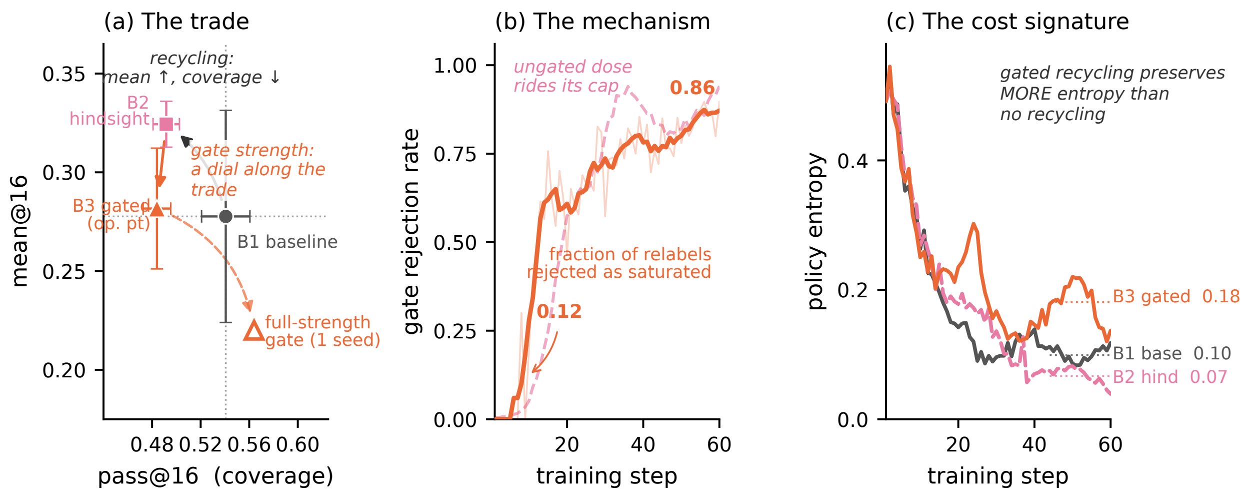

Figure 8: **Recycling-induced sharpening and its derived mitigation (Countdown, tier 1).** *(a)* Three seeds, both axes: recycling (B2) buys mean@16 with pass@16 coverage. On this tier the moderate gate (B3) gives back most of the mean without moving coverage — tier-1 destinations saturate early, so admitted relabels are few — while full-strength gating (corrected decay, 1 seed) returns coverage *above* baseline at the mean's full price: gate strength is a dial along the trade, and its useful middle lives on the frontier tier, where the same moderate setting restores coverage to baseline (.279±.019 vs. .274) keeping ∼60% of the mean gain (§6.8). *(b)* The mechanism in-run: the gate's rejection rate rises monotonically as the model's reachable set saturates its relabel destinations, while the ungated dose rides its cap. *(c)* Gated recycling preserves *more* policy entropy than no recycling — admitted relabels are a spread-preserving gradient.

an alternative reading of the regression that a uniform-with-replacement control cell would separate; it is queued with the replication seeds. Beyond the predictions: the teacher's posterior *learns* real difficulty from ∼1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$) but its allocation is *posterior-starved* — 3,200 draws over 7,473 prompts leave Thompson sampling near-uniform, and seed 2 shows that starvation can neutralize the treatment itself. Two posterior diagnostics sharpen this. *The map is real:* two independent teacher runs' posteriors agree with each other (cross-run Spearman ≈.45 on co-visited prompts, n≈900, ≈ .67 self-consistency with a shared latent) three times better than either agrees with external 7B-model difficulty annotations (.10–.22) — the teacher learns *this model's* difficulty landscape, reliably, from ∼1.25 visits per prompt; the published $\rho = -0.17$ against the 7B key understates what the posterior knows. *But it is compressed:* across external-difficulty quintiles spanning solve rates .33 → .98, mean $\hat{p}$ moves only .076 → .103 — Thompson sampling sees a nearly flat utility landscape, which *is* the allocation-starvation mechanism. A sharpening note in a different currency: response length separates the runs by estimator, perfectly — all three GRPO runs compress to 209.6±1.8 tokens while both MaxRL runs sit at 232.4±5.7 (common ∼365-token warmstart; the teacher moves length ≤ 3 tokens): the estimator, not the curriculum, sets how hard the policy templates its outputs, visible even where token entropy is ambiguous. Small model; the pre-registered outcomes were orderings and signs on the registered run.

Takeaway. The maze main effect stands on nine runs; the LLM-scale interaction is 1-of-2-seeds and tracks steering intensity — the decisive experiment is a steering-controlled multi-seed cell, queued.

6.8 What does recycling cost? (Countdown 2×2, pre-registered): two nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool's simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §6.3's weight-induced collapse — data-induced rather than weight-induced, and to our knowledge unreported (no prior work measures pass@k under hindsight in RLVR). Our algebra predicts it: a relabel to a self-achieved value is a task at $p \approx 1$, where $u(p)$ says training signal is zero — the softmax pays for those updates out of the exploration tail. *Takeaway: recycling needs an admission rule, and the estimator already provides it — the same derived utility that schedules sampling should gate the relabel destination.*

6.9 Does the gate recover it? (Countdown v2 pool, 3 seeds, pre-registered). We rebuilt the pool with real structure (random operand permutation, parenthesization, exact division; a headroom gate excluded guesser-saturated tiers) and ran three arms × three seeds: no recycling (B1), ungated recycling (B2), gated recycling (B3). Setup note with a theory reading: recycling has a *syntactic prerequisite* — the pre-SFT base model produces zero verifier-valid failures (relabel yield 0.000, n=384), so there is nothing to recycle until SFT installs the output format; post-SFT the model is a near-independent guesser whose pass@16 sits at a tier-invariant ∼75% of the independence bound $1 - (1 - \text{pass@1})^{16}$ (ratios .744/.783/.755 across tiers) — the baseline our headroom gate measures against. **Sharpening replicates:** at tier 1, recycling buys mean@16 +.046 (.278±.054 → .324±.012) and pays −.049 of pass@16 (.541±.020 → .492±.011), both outside seed noise; the frontier tier moves the same direction. **The timing is dose-then-damage:** recycling's frontier-tier mean gain is installed by step 30 — +.014±.008 paired, all seeds positive, at only ∼28% of the eventual relabel dose — while the coverage cost only materializes by step 45 (−.048±.019 paired, *all seeds negative*, after ∼65% of dose): the benefit precedes the damage, so early-stopping on mean-accuracy validation would bank the gain and never see the bill. The damage is also visible on the training side: the ungated arm ends with more unsolvable-at-16 frontier training prompts than baseline in every seed (paired +.083/ + .063/ + .034). And the 3-seed entropy read sharpens the single-seed one: the *gate's* entropy retention replicates (+.072 paired over baseline, all seeds positive, ∼1.6×), while ungated recycling makes entropy *seed-unstable* (endpoint spread ±.062, 4–9× the other arms') — consistent with its noisy frontier-coverage endpoints. **The gate works where the algebra says it should:** at the frontier tier (destinations not yet saturated) the gated arm returns coverage to baseline (.279±.019 vs. .274±.015) while keeping ∼60% of the mean gain; at tier 1 (destinations saturate early, so the gate blocks nearly all relabels) it neutralizes recycling almost entirely. Implementation disclosure: a decay-math bug made all three-seed gated runs gate *weaker* than designed (effective destination threshold $\hat{p} \approx 0.64$ where the design implies 0.92), so the reported operating point is an under-gated one; the corrected-decay rerun (1 seed) gates ∼3× harder and pushes tier-1 coverage *above* baseline at the mean's full price (Fig. 8a). Together the three arms trace a monotone dose–response — no gate: full mean gain, coverage lost; under-gated: ∼60% of the gain, coverage restored; full-strength: gain erased, coverage above baseline — `gate_max_p` is a dial along a mean-vs-coverage frontier, and the bug, favorably for the method, sampled its middle. In-run telemetry shows the mechanism (Fig. 8b–c): rejection rises .12 → .85 as the reachable set saturates, and the gated arm ends with *more* policy entropy than the no-recycling baseline. A pre-registered `gate_max_p` dose sweep at fixed decay is the standing follow-up.

Takeaway. Sharpening is a reproducible property of exact-verifier recycling, and the derived gate converts it into a tunable mean-vs-coverage trade whose useful range sits where destinations are not yet saturated.

7 Related work

The coverage tension. RLVR itself trades coverage for precision: pass@1 rises while pass@k at large k falls below the base model [4], with entropy collapse as the mechanistic signature [6] and overtraining on saturated problems as a locus [5]. Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator's implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as the *cure* for coverage loss [7]; our GRPO arms show the cure is estimator-conditional — those studies never varied the estimator. The estimator–difficulty interaction itself is a heating 2026 theme — SC-SDPO [3] shows GRPO's normalization implicitly encodes a difficulty weighting, and A-GRAE [24] redesigns the advantage around difficulty focus — but no prior work runs *identical* curricula under two estimators or measures the resulting pass@k sign flip.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by $\hat{p}(1-\hat{p})$, which Prop. 1's corollary identifies as the $N{=}2$ slice of our derived utility. Reinforce-Ada [13] — the adaptive sampler MaxRL itself points to — adapts the rollout count *within* a prompt until a signal criterion is met, holding the prompt distribution fixed; our teacher is the complementary move, reallocating *across* prompts at fixed group size, and Prop. 3 is the idealized form of both (water-filling spends its next rollout wherever $p(1-p)^N$ is largest, across or within). DAPO's dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward pass@k but keep uniform sampling. **Hindsight and self-training for LLMs.** Exact-verifier relabeling has precedent outside RLVR loops — CodeIt [19] (ARC program synthesis, relabel + prioritized replay) and proof-tree relabeling [20] — and LLM-judged variants relabel agent trajectories (AgentHER, HSL). Concurrent 2026 work brings relabeling-adjacent mechanisms into RLVR-style training: LfH [18] (a VLM proposes one shared hindsight instruction per failed group for VLA post-training; the relabeler is a judge, their stated limitation) and ZipRL's hindsight response replay [21] (densifying GRPO signal for multi-turn agents). None of these measures pass@k under relabeling — precisely the currency where we find recycling's hidden cost. The nearest phenomenon report is in same-problem rejection finetuning: arXiv:2606.07856 [22] document a STaR-trained model winning pass@8 while the base model overtakes at pass@256 — concentration over reach — for SFT on a model's own verified outputs. Recycling-induced sharpening is the cross-task, on-policy-RL sibling, and our gate is its estimator-derived admission rule; gating RLVR updates by pass-rate-like saturation signals has independent precedent for *vanilla* updates [23], not for relabel destinations. So the precise claims we retain: no prior work (i) derives the sampling utility from the estimator's own expected signal, (ii) couples it with success-conditioned weights it provably matches, or (iii) *measures the coverage cost of hindsight relabeling in RLVR* and gates relabel admission by destination saturation.

8 Limitations and negative results

Scale. The LLM-scale teacher×estimator interaction is 1-of-2 seeds and tracks steering intensity: the replication seed of the GRPO+teacher cell did not reproduce the registered regression, and its telemetry shows the treatment was barely delivered (§6.7). The GSM8K teacher is also coverage-neutral-to-negative under *both* estimators at this budget, so the contrast there concerns the sign of the mean-accuracy effect, not a coverage rescue. The maze estimator main effect (nine runs, zero exceptions) is the safety result this paper stands on; a steering-controlled multi-seed LLM cell is the decisive next experiment.

Scope of the surrogate. Advantage mass is not gradient norm, SNR, or expected improvement (Remark 1), and §6.5 exhibits the regime where reallocating it buys nothing (learnable-everywhere pools). Recycled-gradient exactness is a property of the environment's relabel map; noisier verifiers will land between our fixed-pool and one-shot endpoints (+0.22 vs. +0.01 AUC).

What did not transfer. A 4× duration run refuted our own duration hypothesis (the deep-frontier stall is a per-step-competence ceiling; depth needs capacity, not schedule); γ-concentration did not transfer from chains to the maze, as the compounding model predicted; adaptive truncation order underperformed fixed T; and feeding relabeled successes to the teacher's posterior inflates it, so the posterior sees requested-task evidence only.

Corrections along the way. Two early claims were retracted on internal review — a teacher-steering read traced to an epoch-frozen sampler (the fixed run confirmed the effect), and a "beats the oracle" comparison traced to a floor handicap in the oracle arm (§6.1 reports the corrected tie). Full postmortems live in the repository's audit trail.

Reproducibility

Every proposition has a Monte-Carlo verification script; every figure in this paper regenerates from JSON artifacts committed to the repository (`paper/figures/`, with LLM-run endpoints vendored in `paper/figures/data/`); the documentation passed a two-round adversarial audit. Known gaps, stated rather than papered over: the IsaacLab pilot is reported from an independent team's summary (raw logs live with that team), and per-step timing fractions come from verl's in-run telemetry in the ray session logs. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.

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